In the case of an election, for example, the sample space is the set of all candidates eligible to be elected. There is a probability distribution (or measure) function, P: $\Omega \rightarrow [0,1]$, where P is a function from the sample space to real numbers in the interval from 0 to 1. You could plug in P(*candidate A*) and it will spit out a number between 0 and 1 indicating the probability of candidate A winning the election.

Note

Probability theory is more complicated than what we've articulated here and involves a branch of mathematics called *measure theory*. For our purposes, we do not need to delve any deeper into probability theory than we already have. We will stick with an informal and mathematically nonrigorous introduction to the probability concepts we need.

The *support* of a probability distribution is another term we will use. The support is just the subset of outcomes that are assigned nonzero probabilities. For example, temperatures can't be less than 0 Kelvin, so negative temperatures would be assigned probability 0; the support of the probability distribution over temperatures would be from 0 to positive infinity. Since we generally don't care about outcomes that are impossible, you'll often see "support" and "sample space" used interchangeably, even though they may not be the same.

7.2.1. Priors and posteriors

If we were to ask you "what is the probability of each candidate in a 4-way race winning?" without specifying who the candidates were or what the election was about, you might refuse to answer, citing insufficient information. If we really pressed you, you might say that since you know nothing else, each candidate has a ¼ chance of winning. With that answer, you've established a *prior probability distribution* that is uniform (each possible outcome has the same probability) over the candidates.